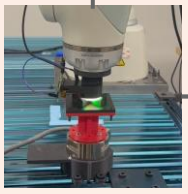
(a) Real contacts

Labels need calibration and verification

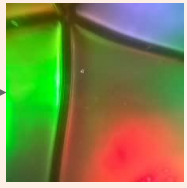
Contact state from the robot pose

GT pose q^*

Mesh at q^*



Registered press

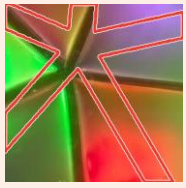


Real image



GT depth

after re-placing the object



old calibration

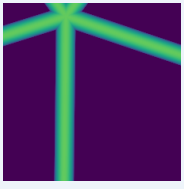


re-calibrated

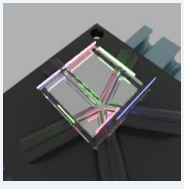
(b) Simulate contacts

Synthetic tactile images need sensor-specific modeling

GT pose q^* +



GT depth



Simulated contact

render



PBR

or



Learned renderer

(c) AlignTac: real appearance + aligned geometry

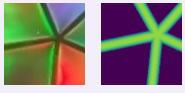


Initial estimate q_0

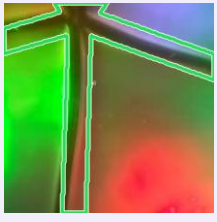
unlabeled real image + known mesh

AlignTac

train



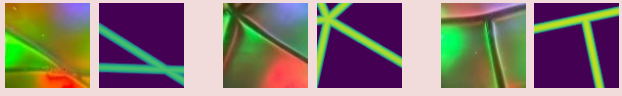
few labeled contacts



After alignment



Pseudo-label



Many pseudo-labeled contacts for supervision